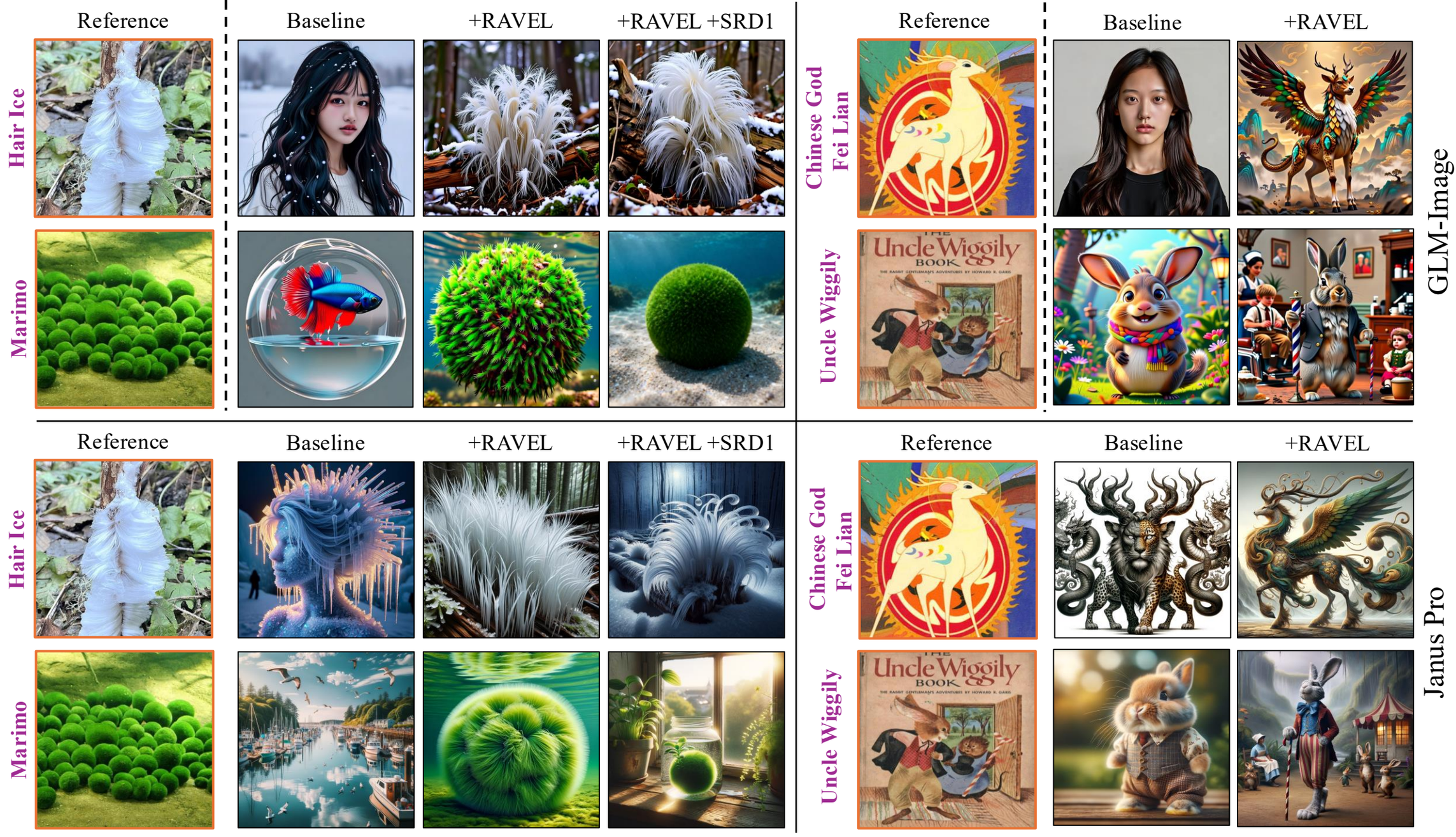

Figure R1. RAVEL generalizes to autoregressive T2I models. Qualitative comparison on GLM-Image and Janus Pro across four rare concept categories: a natural phenomenon (Hair Ice), a biological rarity (Marimo), a mythological entity (Chinese God Fei Lian), and a literary character (Uncle Wiggily). Baseline models exhibit characteristic failure modes — semantic collapse (Hair Ice rendered as a girl with black hair; Marimo as a marina harbor or fish bowl), generic anthropomorphism (Fei Lian rendered as a plain portrait or a generic winged beast without its distinctive leopard-deer hybrid anatomy), and narrative incoherence (Uncle Wiggily rendered as a generic cartoon rabbit without anthropomorphic clothing or cane). RAVEL consistently recovers concept-specific attributes across both architectures, demonstrating that structured KG grounding improves rare concept fidelity independently of the generative backbone.